Sedimentation in a box

Exercise sheet — Journal club

Model. A colloidal sphere sits in a vertical cell of height L, with the floor at $x = 0$ and the ceiling at $x = L$. Its buoyant weight mg (m is its mass minus that of the fluid it displaces) pulls it down against Stokes friction γ, and the fluid kicks it about thermally. In the overdamped limit,

$$\mathrm{d}x = -v\,\mathrm{d}t + \sqrt{D}\,\mathrm{d}W, \qquad v = \frac{mg}{\gamma}, \qquad D = \frac{2k_BT}{\gamma}.$$

Throughout,

$$h = \frac{D}{2v}, \qquad k_n = \frac{n\pi}{L}.$$

Convention: as in the talk, $B = D$ is the variance per unit time, $\langle(\Delta x)^2\rangle = D\,\Delta t$, so the ordinary diffusion coefficient is $D/2 = k_BT/\gamma$ (Einstein). Parts (a)–(d) are the core exercise; part (e) is an optional extension. Equation numbers refer to Gardiner, 2nd edition.

(a) The closed cell: find the stationary state

The walls are hard.

1. Write the forward equation and the current J, and the conditions at the walls (5.2.13).
2. Find the stationary density $p_s(x)$ from the potential solution (5.2.39), and normalise it. Show that its decay length is $h = k_BT/mg$. Why does the friction γ drop out?
3. Estimate h for a silica sphere of radius $0.5\,\mu$m in water, with density difference 1.0 g/cm^3, at room temperature.
4. Find the mean height $\langle x\rangle_s$, and check the limits $L \ll h$ and $L \gg h$.

Hint: $k_BT \approx 4.0 \times 10^{-21}$ J at 293 K. Counting grains at different heights in this profile is how Perrin measured Avogadro's number (§§5.2.1, 5.2.2).

(b) The closed cell: find the modes

1. Solve the backward eigenproblem, with the reflecting condition (5.2.69) at both walls,

$$-v\,Q' + \tfrac{1}{2}D\,Q'' = -\lambda Q, \qquad Q'(0) = Q'(L) = 0.$$

   Try $Q = e^{ax}$, and show that $a = 1/2h \pm ik$ with $\lambda = v^2/2D + \frac{1}{2}Dk^2$. Then show that

$$Q_n = e^{x/2h}\Big[\cos k_n x - \frac{\sin k_n x}{2hk_n}\Big], \qquad \lambda_n = \frac{v^2}{2D} + \frac{n^2\pi^2 D}{2L^2}, \qquad n = 1, 2, \ldots,$$

   together with $\lambda_0 = 0$, $Q_0 = 1$. Are there any other eigenvalues?

2. Write down $P_n = p_s\,Q_n$, and check that it carries no current through the walls.

3. Interpret the two terms of the spectral gap λ_1. Which one dominates in a tall cell, $L \gg h$?

4. A particle released at the ceiling of a tall cell, say $L = 20h$, needs a time of about L/v to settle. Compare this with $1/\lambda_1$. How can the density relax toward p_s faster than the particle can get to where p_s lives? Resolve the paradox.

Hints: For other eigenvalues, try a real a (so $\lambda < v^2/2D$). For the paradox, look at the size of each term of the expansion (5.2.88), $p(x, t \,|\, x_0, 0) = \sum_\lambda P_\lambda(x)\,Q_\lambda(x_0)\,e^{-\lambda t}$, for $x_0 = L$ and x near the floor.

(c) Make the walls sticky

Now coat both walls so that a particle that touches either one sticks for good. Both walls become absorbing: $p = 0$ there (5.2.14), and $Q = 0$ in the backward problem (5.2.67).

1. Why is there no stationary state now?
2. Find the backward eigenfunctions Q_n and the rates λ_n. Compare with (b): which eigenvalue is lost, and which survive?
3. Show that $P_n = p_s\, Q_n$ still solves the forward equation, with p_s the barometric profile of (a), even though p_s is no longer a stationary state (5.2.91). Normalise the pair.
4. Show that the QSD is
$$q(x) \propto e^{-x/2h} \sin \frac{\pi x}{L},$$
and normalise it. Sketch q and p_s for $L = 8h$, and find the height of the peak of q. The survivors' profile decays over $2h$, not h, so they sit higher than particles in equilibrium. Why?
5. The survival probability decays as $e^{-\lambda_1 t}$. At what rate does p/S approach the QSD? Show that it does not depend on v, and explain why.

Hint: $\displaystyle\int_0^L e^{-x/2h} \sin k_1 x \, \mathrm{d}x = \frac{k_1 \left(1 + e^{-L/2h}\right)}{(2h)^{-2} + k_1^2}$ (§§5.2.5, 5.2.7).

(d) Where and when does it stick?

1. Show that a particle released at x_0 sticks to the ceiling with probability

$$\pi_L(x_0) = \frac{e^{x_0/h} - 1}{e^{L/h} - 1}.$$

   Use (5.2.186)–(5.2.190). For $L \gg h$, how close to the ceiling must it start to have a fair chance?

2. Find the mean time to stick, $T(x_0)$, from (5.2.154), and show that $T = [x_0 - L\,\pi_L(x_0)]/v$. What does it reduce to when $L \gg h$, away from the ceiling?

3. Now release particles distributed as the QSD. Without integrating anything, show that the time to stick is exponential with mean $1/\lambda_1$. Find the currents through the floor and the ceiling, check that they add up to λ_1, and find the fraction of particles that stick to the ceiling.

4. What do the results of 3 say about $\int_0^L q\,T\,\mathrm{d}x$ and $\int_0^L q\,\pi_L\,\mathrm{d}x$?

Hints: Here $\psi(x) = \exp\int^x 2A/B\,\mathrm{d}x' = e^{-x/h}$. Gardiner's printed (5.2.189)–(5.2.190) have ψ where $1/\psi$ belongs. In (5.2.154), x/v is a particular solution. For 3, what happens to $p(x,t)$ if it starts as the QSD? (§§5.2.7, 5.2.8)

(e) Optional: why the spectrum survived

In (c) the walls changed completely, from no current to no density, yet every nonzero rate stayed the same.

1. Show that when A and B are constant, the current $J(x,t)$ of any solution obeys the same forward equation as p.
2. In the closed cell, what does J satisfy at the walls? Conclude that the current of each closed-cell mode is a sticky-cell mode with the same rate. Check it: with $Q_n^{\text{sticky}} = e^{x/2h}\sin k_n x$, and both pairs normalised so that $\int P_n Q_n \,\mathrm{d}x = 1$, show that $J[P_n^{\text{closed}}] = \frac{1}{2} D k_n \, P_n^{\text{sticky}}$. What happens to the mode λ_0?
3. Conversely, let $\tilde{p}$ be a sticky-cell mode with rate $\lambda > 0$. Show that $P = \tilde{p}\,'/\lambda$ is a closed-cell mode with the same rate, and that its current is $\tilde{p}$. So the two spectra differ by exactly one eigenvalue, $\lambda_0 = 0$.
4. Where did you use that A is constant? What equation does J obey when the drift is $A(x) = -U'(x)$?

Hint: Differentiate $J = Ap - \frac{1}{2} D\,\partial_x p$ with respect to t, and use $\partial_t p = -\partial_x J$.

The barometric profile is the potential solution (5.2.39) with constant drift; the sticky cell is additional material beyond the book examples. References: J. Perrin, *Mouvement brownien et réalité moléculaire*, Ann. Chim. Phys. (8) **18**, 5 (1909); S. Chandrasekhar, *Stochastic problems in physics and astronomy*, Rev. Mod. Phys. **15**, 1 (1943), chapter II, for sedimentation as a diffusion problem. The accompanying key uses the same part labels.